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Programmazione Concorrente e Distribuita

Ingegneria e Scienze Informatiche - UNIBO
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[Lab Notes]

GUI FRAMEWORKS & CONCURRENCY

GUI FRAMEWORKS & CONCURRENCY

- Once upon a time GUI applications were single-threaded...
  - GUI events processed by a “main event loop”
- ..modern GUI Frameworks are not so different
  - creating a dedicated **event dispatch thread** (EDT) for handling GUI events
  - the thread fetches events off a queue and dispatches them to application-defined event handlers
- Most of the modern GUI Frameworks are **single-threaded**
  - Java, QT, NextStep, Mac OS Cocoa, XWindow...
- Multithreaded GUI as a “failed dream” [*]
  - many attempts, failed due to the generation of problems with race conditions and deadlock

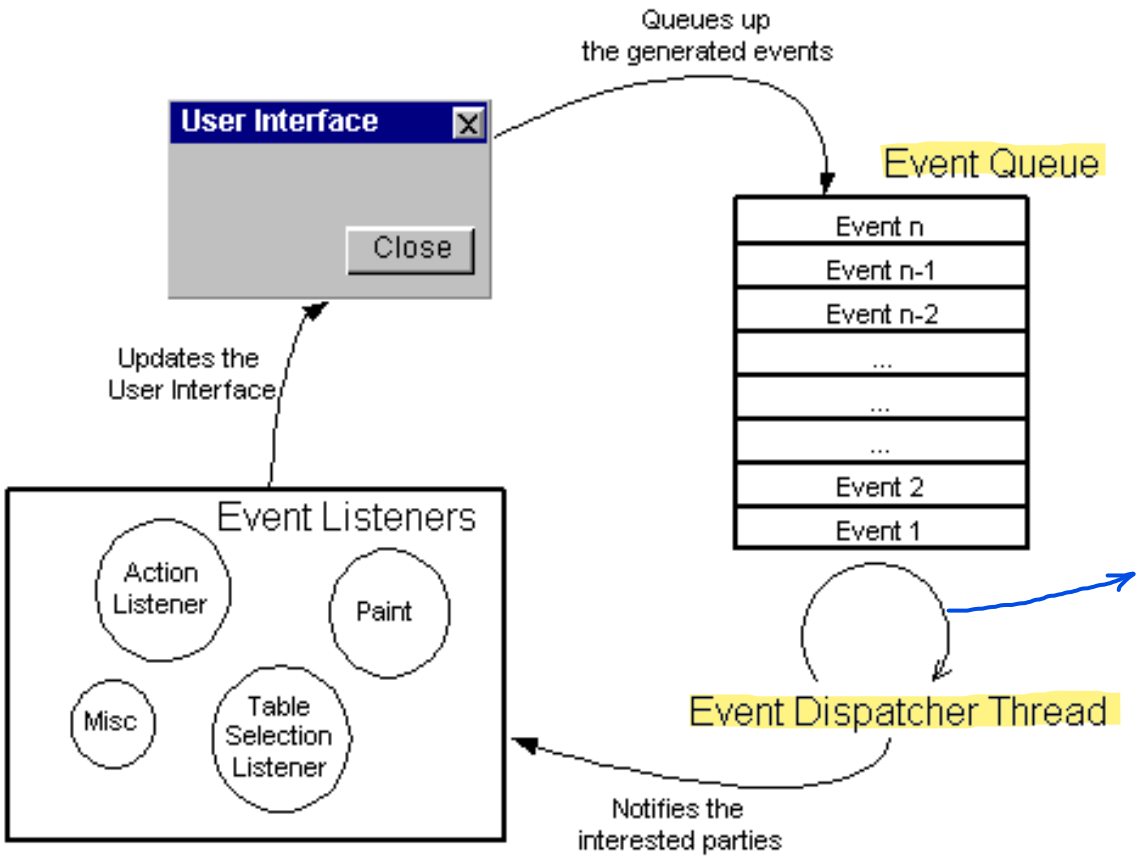
[*] referred in this way by Sun VP Graham Hamilton in his blog https://community.oracle.com/blogs/kgh/2004/10/19/multithreaded-toolkits-failed-dream [NO MORE AVAILABLE]

MULTITHREADED GUI FRAMEWORKS: THE PROBLEM

- Clash between input event processing and OO modeling of GUI components
  - that can easily lead to *deadlocks* and *race conditions*
- Inconsistent lock ordering
  - in managing user-initiated actions and application-initiated actions
    - user actions “bubble up” from OS to the application
      - OS mouse click ->mouse click event of the GUI toolkit -> high-level event of the application listener...
    - application-actions “bubble down” from application to action
      - changing background color of a component at the application level -> dispatched to component class -> dispatched to OS for rendering,,,
    - most of the actions need to lock objects
  - more generally related to MVC implementations

SINGLE-THREADED GUI: EVENT-LOOP ARCHITECTURE:

I componenti grafici non sono Thread-Safe perchè, nella filosofia di design, chi può chiamare il metodo per cambiare lo stato di quel componente è sempre sotto ad un EDT



esegue tutti gli eventi presenti nella coda (Quando un evento arriva, il dispatcher thread lo preleva dalla coda e chiama il gestore dell'evento (Handler/Listener) associato. Il codice all'interno del listener/handler viene eseguito da questo thread)

Unico Thread che è in grado di cambiare lo stato di componenti grafiche di GUI

[Figure taken from: https://www.javaworld.com/article/2073477/swing-gui-programming/customize-swingworker-to-improve-swing-guis.html]

SINGLE-THREADED GUI: CONSTRAINTS

- Achieving thread-safety via **thread-confinement**
  - *all GUI objects are accessed exclusively by the event thread*
    - including visual components and data models
  - the application developer must make sure that these objects are properly confined
- Sequential event processing
  - events like kind of task to be processed sequentially by the event thread
- Problems and challenges
  - if one task takes long time to execute, other task must wait
    - blocking the overall GUI
  - > ...so tasks that execute in the event-thread must return quickly
  - to initiate a long-term task a separate thread must be used
    - es: spell-checking a document, searching the file system
  - > ...but typically a long-term task must provide a visual feedback for indicating progress or when it completes
    - and this code need to be executed by the event thread...

THREAD CONFINEMENT IN SWING

- All Swing components (such as JButton and JTable) and data models (e.g. Table Model and Tree Model) are confined to the event thread
  - any code that access these objects must run in the event thread
- Some exceptions
  - Swing methods that can be safely called from any thread
    - clearly identified in the Javadoc as thread-safe

THREAD-SAFE SWING METHODS

Tutte le volte che devo modificare il codice o ottenere/lavorare con i componenti non solo metto questa parte nella view, ma lo devo far fare al EDT (perchè se no ho possibilità di Corse Critiche)

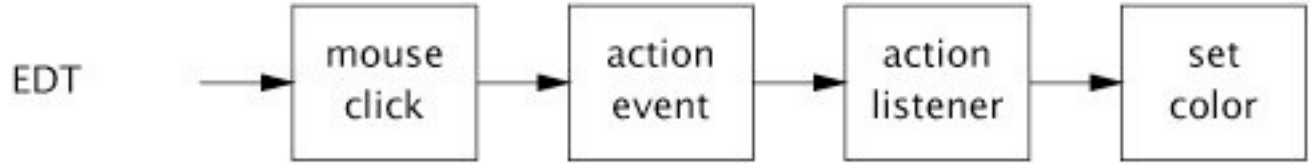
- Thread-Safe methods
  - **`SwingUtilities.isEventDispatchThread`**
    - to check if the current thread is the event thread
  - **`SwingUtilities.invokeLater`** Concettualmente, mette un task da fare nella Event Queue
    - to schedule a `Runnable` for execution on the event thread
  - **`SwingUtilities.invokeAndWait`** Comportamento sincrono: quando chiamato, viene accodato il task in Event queue ed aspetto che sia completato da EDT
    - to schedule a `Runnable` task for execution on the event thread, blocking the current thread until it completes
    - cannot be called by the event thread
  - methods to enqueue and repaint or revalidation request on the event queue
  - methods for adding or removing listeners
    - can be called from any thread, but listeners will always be invoked in the event thread

SHORT-RUNNING GUI TASKS

- Can be executed directly by the event thread
- Simple example

```
final Random random = new Random();
final JButton button = new JButton("Change Color");
...
button.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        button.setBackground(new Color(random.nextInt()));
    }
});
```

- The control never leaves the event thread
  - the event originates in the GUI toolkit, is delivered to the application, the application modifies the GUI in response to user's action



LONG-RUNNING TASKS

- Some of the processing must be *offloaded* to another thread
  - a pool thread, or created on-the-fly (warning)
  - exploiting executors (next modules)
- Two main cases
  - long-term task without visual feedbacks
    - simple case, quite unfrequent
  - long-term task with visual feedbacks
    - complex case, most frequent

BINDING A TASK WITHOUT VISUAL FEEDBACKS

- Naive solution: creating a new thread - simple example:

```
button.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        new Thread(()->{ doBigComputation(); }).start();
}});
```

- This solution could be problematic
  - more robust solution consider offloading the task to do to a thread pool (as a master worker solution)
    - to avoid the cost of thread creation
    - to limit the number of threads to be used

LONG-RUNNING TASK WITH USER FEEDBACKS

- The long-running task must submit another task to run in the event thread whenever the user interface must be updated
- Example (toy, to be refactored):

```
button.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        button.setEnabled(false);
        label.setText("busy");
        new Thread(()-> {
                try {
                    doBigComputation();
                } finally {
                    SwingUtilities.invokeLater(() -> {
                            button.setEnabled(true);
                            label.setText("idle");
                    });
                }
            }).start();
    }
});
```

TOWARDS A CLEANER DESIGN

- Revisiting the Concurrent Model View Controller
- Mapping
  - encapsulating control in agents
    - active components of the program
    - encapsulating the control logic
  - model as shared passive components
    - can be used by agents
    - encapsulating mutex properties
    - includes coordination media
  - view as shared passive components **+ one active component**
    - passive components
      - enabling and mediating the interaction with users
      - observed and used by agents
    - active component (EDT)
      - responsible of executing any tasks about the GUI

SwingWorker (ARGUABLE DESIGN)

- Java 6.0 provides auxiliary classes for making it easier to program complex long-term tasks that can interact with the GUI
  - a
- `SwingWorker` class
  - provide a direct support for task cancellation, completion notification and progress indication

```
class SwingWorker<T,V> implements RunnableFuture<T> {
  ...
  // to be overridden
  protected abstract  T doInBackground();
  protected  void     done()

  protected final void publish(V... chunks)
  protected void process(List<V> chunks);
  ...
  // to be directly used
  boolean cancel(boolean mayInterruptIfRunning);
  protected  void setProgress(int progress);
  ...
}
```

TASK EXECUTION AND INTERFACE UPDATE (1/2)

- Support for asynchronous task execution & consequent interface update
  - `doInBackground`
    - encapsulate the computational body of the task to be executed asynchronously w.r.t. GUI activity, computing a result or throwing an exception if unable to do so
    - executed by some thread, not by the Swing EDT
  - `done`
    - encapsulate the action to do on the GUI when the task completed
    - executed by the Swing EDT

TASK EXECUTION AND INTERFACE UPDATE (2/2)

- Support for asynchronous update of interfaces
  - `publish(V... chunks)`
    - used from inside doInBackground to deliver intermediate results for processing on the Event Dispatch Thread inside the process method
  - `process(List<V> chunks);`
    - receives data chunks from the publish method asynchronously on the EDT

AN EXAMPLE: SWING WORKER TEST

```
class CounterTask extends SwingWorker<Integer, Integer> {

  protected Integer doInBackground() throws Exception {
    int i = 0;
    int sum = 0;
    int maxCount = 10;
    while (!isCancelled() && i < maxCount) {
      sum+=i;
      i++;
      publish(new Integer[] { i });
      setProgress(100 * i / maxCount);
      Thread.sleep(1000);
    }
    return sum;
  }

  protected void process(List<Integer> chunks) {
    for (int i : chunks)
      System.out.println("Step "+i);
  }

  protected void done() {
    if (isCancelled()){
      System.out.println("Task cancelled.");
    } else {
      System.out.println("Task completed.");
    }
  }
}
```

```java
public class SwingWorkerTest {
  public static void main(String[] args) {
    JTextArea textArea = new JTextArea(10, 20);
    JProgressBar progressBar = new JProgressBar(0, 100);
    CounterTask task = new CounterTask();

    JButton startButton = new JButton("Start");
    startButton.addActionListener(new ActionListener() {
       public void actionPerformed(ActionEvent e) { task.execute();}});

    JButton cancelButton = new JButton("Cancel");
    cancelButton.addActionListener(new ActionListener() {
       public void actionPerformed(ActionEvent e) { task.cancel(true); }});

    task.addPropertyChangeListener(new PropertyChangeListener() {
       public void propertyChange(PropertyChangeEvent evt) {
           if ("progress".equals(evt.getPropertyName())) {
             progressBar.setValue((Integer) evt.getNewValue());
           }
       }
    });

    JPanel buttonPanel = new JPanel();
    buttonPanel.add(startButton);
    buttonPanel.add(cancelButton);
    JPanel cp = new JPanel();
    LayoutManager layout = new BoxLayout(cp, BoxLayout.Y_AXIS);
    cp.setLayout(layout);
    cp.add(buttonPanel);
    cp.add(new JScrollPane(textArea));
    cp.add(progressBar);
    JFrame frame = new JFrame("SwingWorker Test");
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setContentPane(cp);
    frame.pack();
    frame.setVisible(true);
  }
}
```